

Database Management System (DBMS) is software that allows users to create, manage, and manipulate databases efficiently. It provides an interface between users and the database to ensure data is stored, retrieved, and managed properly.

A database is an organized collection of related data. Data can be stored in tables consisting of rows and columns. Each row represents a record, and each column represents an attribute.

The primary goals of a DBMS are data consistency, data integrity, data security, and efficient data retrieval. It minimizes redundancy and maintains accuracy of stored data.

There are different types of DBMS such as hierarchical, network, relational, and object-oriented DBMS. Among these, the relational database model is the most widely used.

In the relational model, data is stored in tables called relations. Each table has a primary key which uniquely identifies each record. A foreign key is used to establish a relationship between two tables.

Structured Query Language (SQL) is used to interact with relational databases. SQL commands are categorized into Data Definition Language (DDL), Data Manipulation Language (DML), Data Control Language (DCL), and Transaction Control Language (TCL).

DDL commands include CREATE, ALTER, and DROP. These commands are used to define and modify database structures.

DML commands include SELECT, INSERT, UPDATE, and DELETE. These commands are used to manipulate data stored in tables.

Normalization is a process used to reduce data redundancy and improve data integrity. It organizes tables into normal forms such as First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF).

First Normal Form eliminates repeating groups and ensures atomic values. Second Normal Form removes partial dependency. Third Normal Form removes transitive dependency.

ACID properties ensure reliable database transactions. ACID stands for Atomicity, Consistency, Isolation, and Durability.

Atomicity ensures that a transaction is completed entirely or not executed at all. Consistency ensures that the database remains in a valid state before and after a transaction.

Isolation ensures that concurrent transactions do not interfere with each other. Durability ensures that committed changes are permanently saved even in case of system failure.

A transaction is a sequence of operations performed as a single logical unit of work. Transactions can be committed or rolled back.

Concurrency control mechanisms are used to manage simultaneous operations without conflicts. Lock-based protocols and timestamp-based protocols are common concurrency control methods.

Deadlock occurs when two or more transactions wait indefinitely for each other to release locks. Deadlock detection and prevention techniques are used to handle such situations.

Indexing improves the speed of data retrieval operations. An index is a data structure that allows faster search operations. B+ trees and hash indexing are commonly used indexing techniques.

Joins are used to combine rows from two or more tables based on related columns. Types of joins include INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN.

Database security involves authentication, authorization, and encryption mechanisms to protect data from unauthorized access.

Backup and recovery mechanisms are essential to restore data in case of hardware failure, software errors, or natural disasters.

Two-phase commit protocol is used in distributed databases to ensure atomic transactions across multiple systems.

Database optimization techniques improve query performance. Query optimization selects the most efficient execution plan for a SQL query.